

A Workshop for Understanding and Using Instrument-free Methods

When Gaussian Copulas Work, When They Crack,
and a Novel ICA Approach for Taking Up the Pieces

Workshop @ YES 26, Rotterdam

24 August 2026

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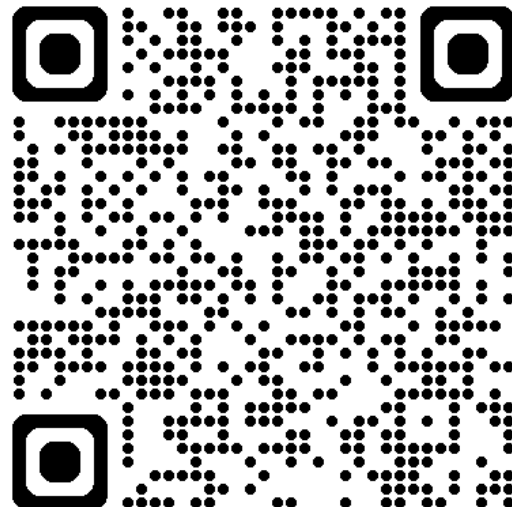
Agenda and Materials

Our plan for today:

- 1) Opening: *5min*
- 2) Introduction to the Gaussian Copula method: *15 min*
- 3) Quick poll/exercise: *5 min*
- 4) Using Gaussian Copula methods in R: *20 min*
- 5) Revisiting the assumptions in a DAG; introducing new method: *10 min*
- 6) *Introducing a new method (ICA) 10 min.*
- 7) Using the new method in R: *10 min*
- 8) *What to do as author / reviewer? 5 min*
- 9) Discussion, Q&A, buffer: *10 min*

Materials:

<https://github.com/HashtagHaschka/Workshop>



Notes/References:

Other stuff you might want or need:

• Papers on Copula:

- Haschka, R. E. (2022). Handling endogenous regressors using copulas: a generalization to linear panel models with fixed effects and correlated regressors. *Journal of Marketing Research*, 59(4), 860-881.
- Haschka, R. E. (2025). Robustness of copula-correction models in causal analysis: Exploiting between-regressor correlation. *IMA Journal of Management Mathematics*, 36(1), 161-180.

Recent paper when Copula cracks:

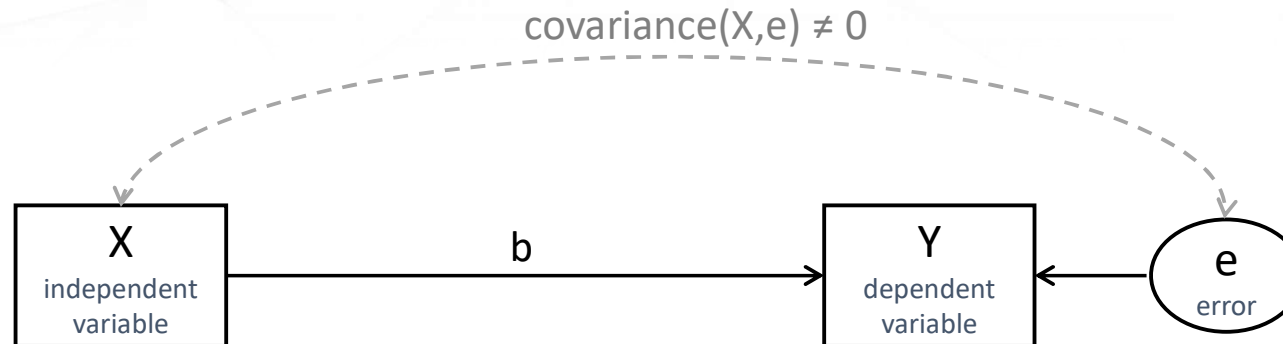
- Dost, Florian and Haschka, Rouven E., The Gaussian Copula Control Function Method Does Not Help Against Traditional Omitted Variable Bias (June 07, 2025). Available at SSRN: <http://dx.doi.org/10.2139/ssrn.5285127>

New instrument-free method for omitted variables bias

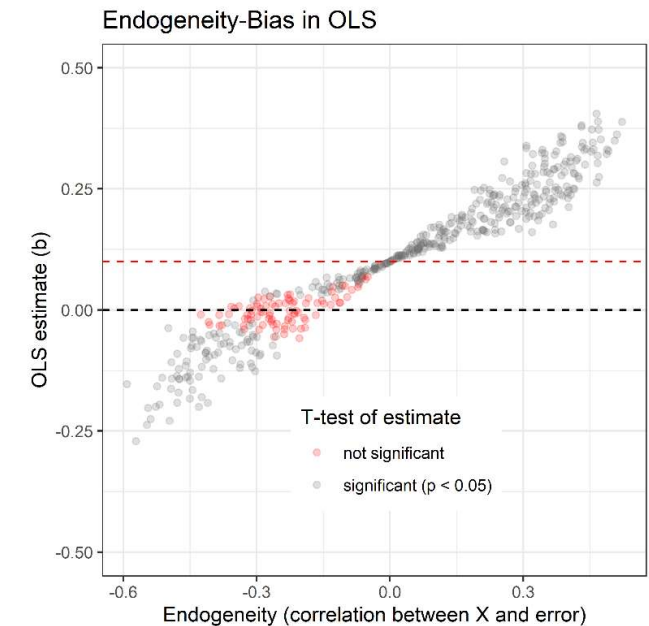
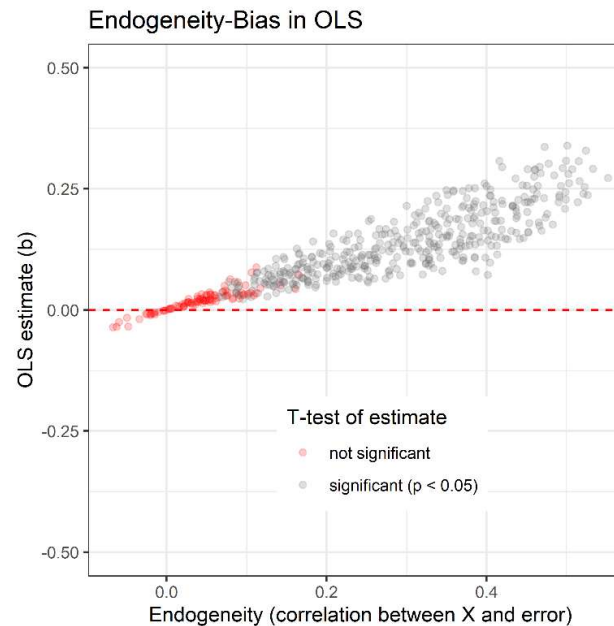
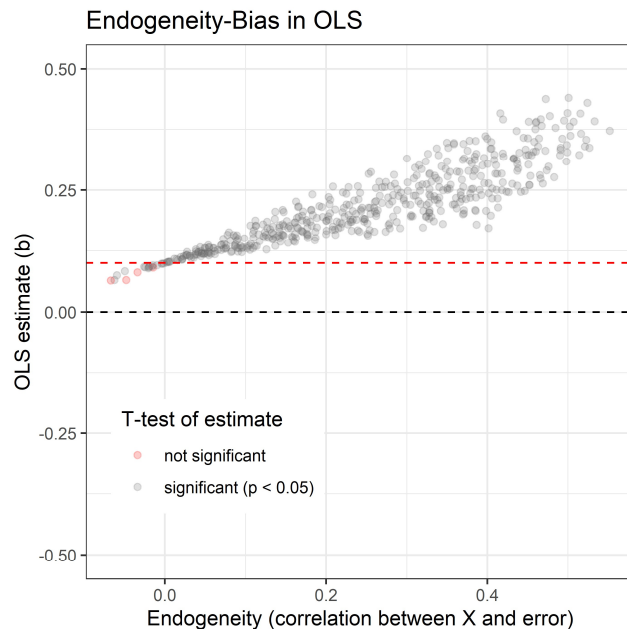
- Haschka, Rouven E. and Dost, Florian, ICA at the Cocktail Party: Casting Instrument-free Omitted Variable Bias Correction as a Blind Source Separation Problem (July 22, 2025). Available at SSRN: <http://dx.doi.org/10.2139/ssrn.5361801>

What is the problem with endogeneity?

What is endogeneity?



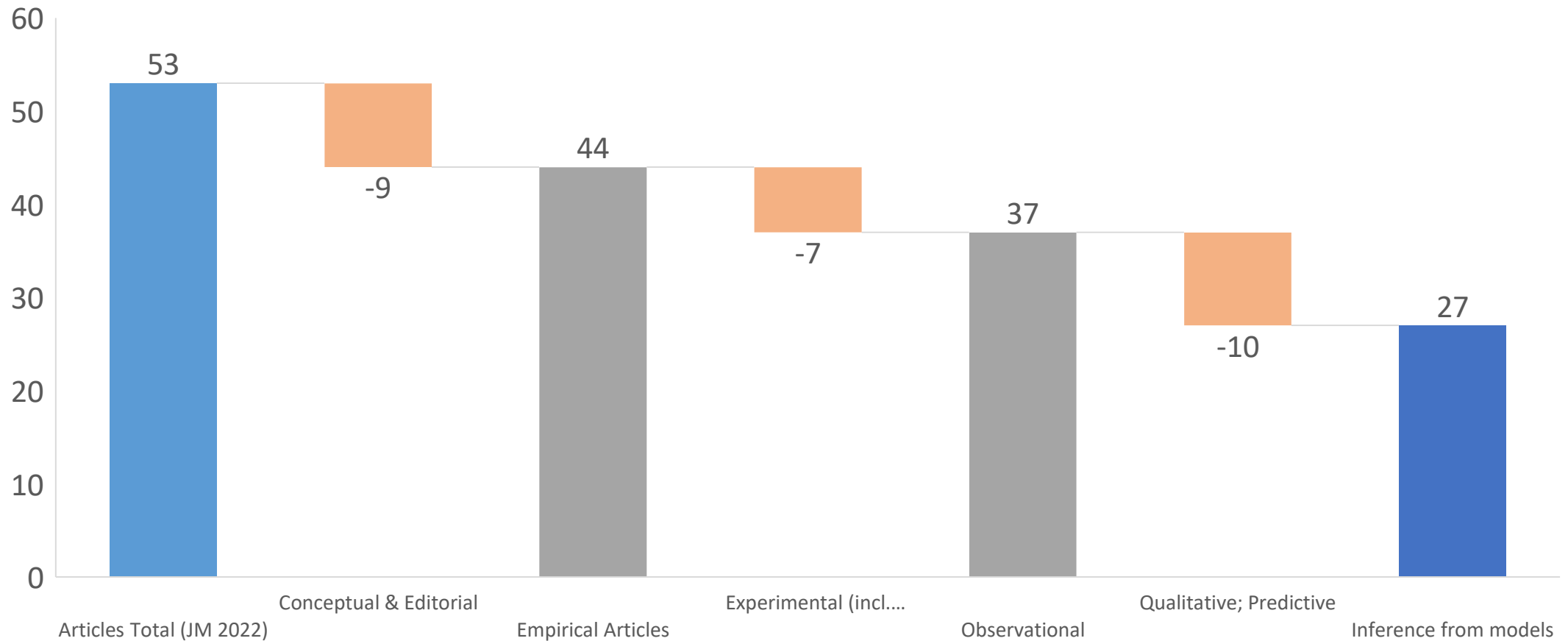
Why does it matter?



Notes/References: Hill et al. (2021) ; own simulations

Who cares about endogeneity?

Articles in the Journal of Marketing, in 2022, by type of included studies



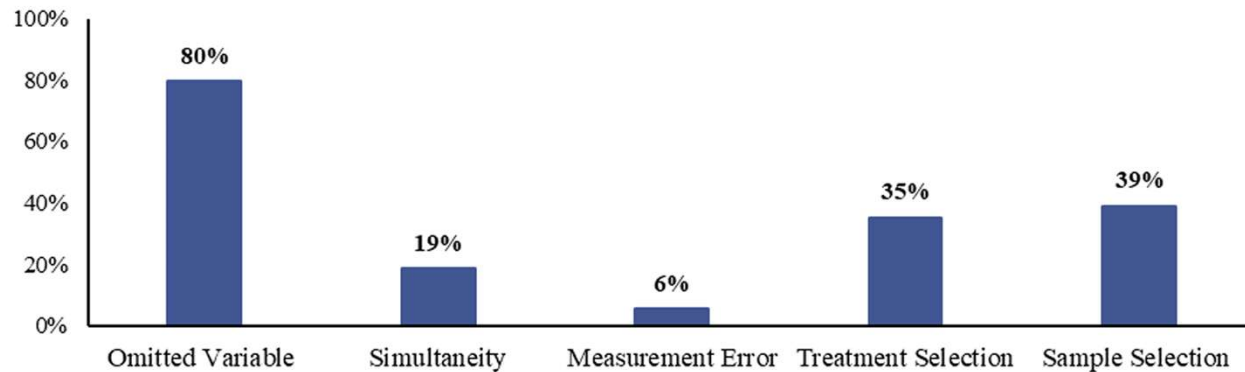
Notes/References: own research, based on all 6 issues of the 2022 Journal of Marketing

Top journals care a lot! Mostly, it is omitted variables bias

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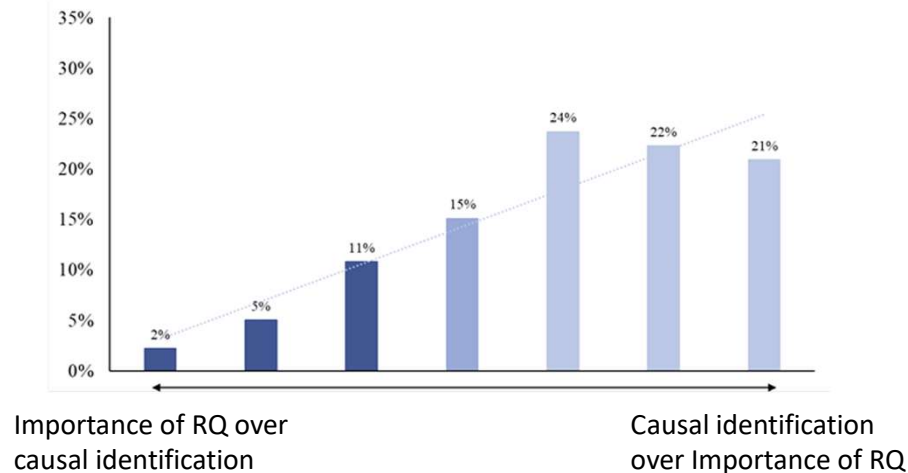
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Sources of Endogeneity Acknowledged



Note: Based on 108 papers with observational data published in 2024 in *JM*, *JMR*, *JCR*, and *MKS*. Multiple categories are possible for substantive focus, theory domain, level of analysis, and sources of endogeneity acknowledged.

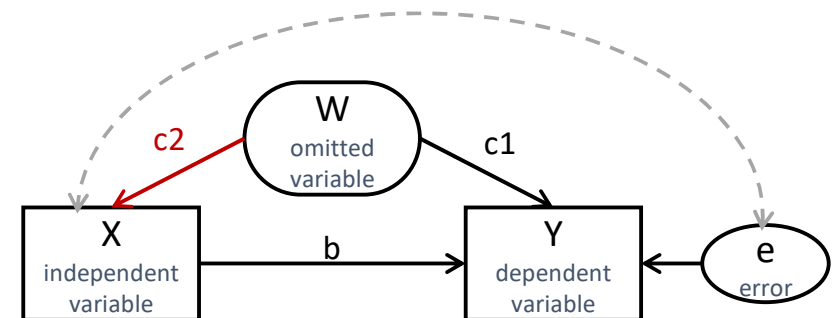
1e: Major Journals Often Prioritize Causal Identification



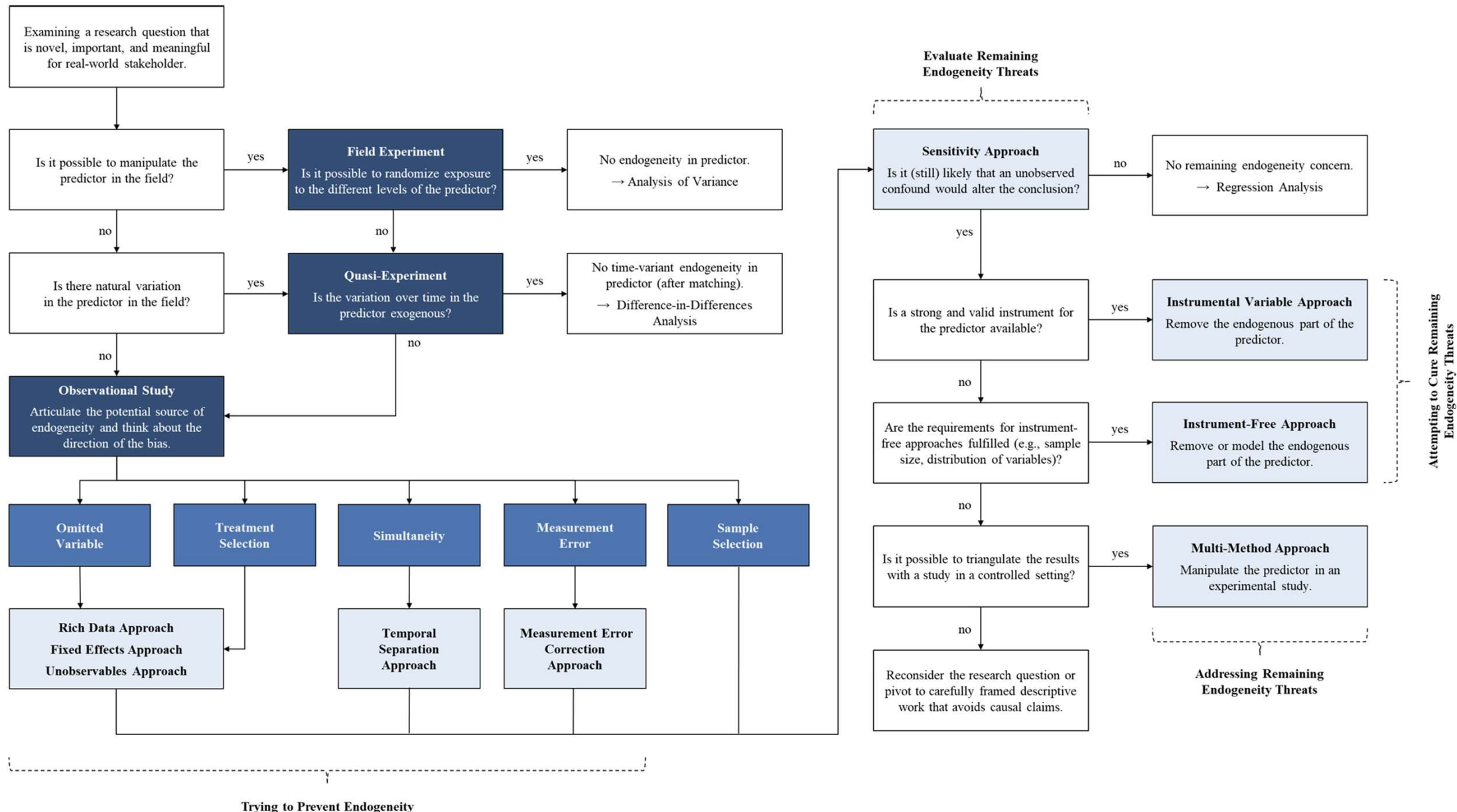
• Omitted variable bias:

Examples:

- Ability affects selection into training, both ability and training affect performance.
- Events (superbowl) increase ad effectiveness; managers plan accordingly; observed ad effects are higher.
- Events (conference) increase hotel demand; managers increase price; price and demand correlate positively.



Notes/References: Hill et al. (2019); Woolridge (2010);
Herhausen, Dennis and Van Heerde, Harald and Ludwig, Stephan and Grewal, Dhruv, Navigating Endogeneity in Marketing Research (March 23, 2026). UNSW Business School Research Paper Forthcoming, Available at SSRN: <https://ssrn.com/abstract=5246466> or <http://dx.doi.org/10.2139/ssrn.5246466>



The Gaussian Copula endogeneity correction is widely used in marketing

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Handling Endogenous Regressors by Joint Estimation Using Copulas

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Journal of the Academy of Marketing Science (2022) 50:46–66
https://doi.org/10.1007/s11747-021-00805-y

METHODOLOGICAL PAPER



Handling Endogenous Regressors Using Copulas: A Generalization to Linear Panel Models with Fixed Effects and Correlated Regressors

Rouven E. Haschka

Addressing Endogeneity Without Instrumental Variables: An Evaluation of the Gaussian Copula Approach for Management Research

Christine Eckert

University of Technology Sydney

Jan Hohberger

University Ramon Llull

Article

Addressing Endogeneity Using a Two-Stage Copula Generated Regressor Approach

Journal of Marketing Research
2025, Vol. 62(4) 601–623
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Sage

Fan Yang, Yi Qian, and Hui Xie

Journal of the Academy of Marketing Science (2025) 53:279–299
https://doi.org/10.1007/s11747-024-01055-4

METHODOLOGICAL PAPER



Dealing with regression models' endogeneity by means of an adjusted estimator for the Gaussian copula approach

Benjamin D. Liengaard · Jan-Michael Becker · Mikkel Bennedsen · Phillip Heiler · Luke N. Taylor · Christian M. Ringle

Journal of Marketing

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Impact Factor: 10.4 / 5-Year Impact Factor: 17.9

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EXPRESS: A Practical Guide to Endogeneity Correction Using Copulas

Yi Qian, Anthony Koschmann, and Hui Xie View all authors and affiliations

Accepted Manuscripts | https://doi.org/10.1177/00222429251410844

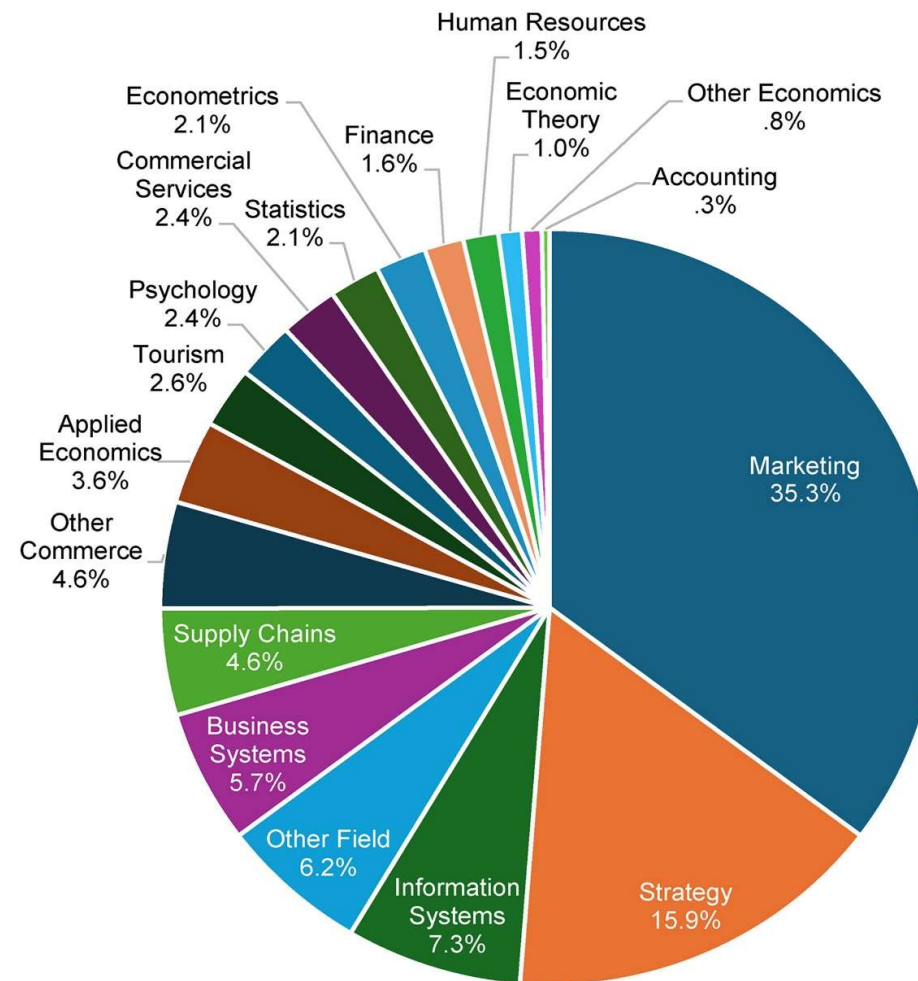
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GCCF use by discipline (from Qian, Koschmann & Xie 2026)

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A: Publications (n = 615) Using Copula Correction by Disciplines



Source: Qian, Y., Koschmann, A., & Xie, H. (2026). A practical guide to endogeneity correction using copulas. *Journal of Marketing*, 90(4), 27-51.

Explicit assumptions for Gaussian Copula Control Function (GCCF) methods

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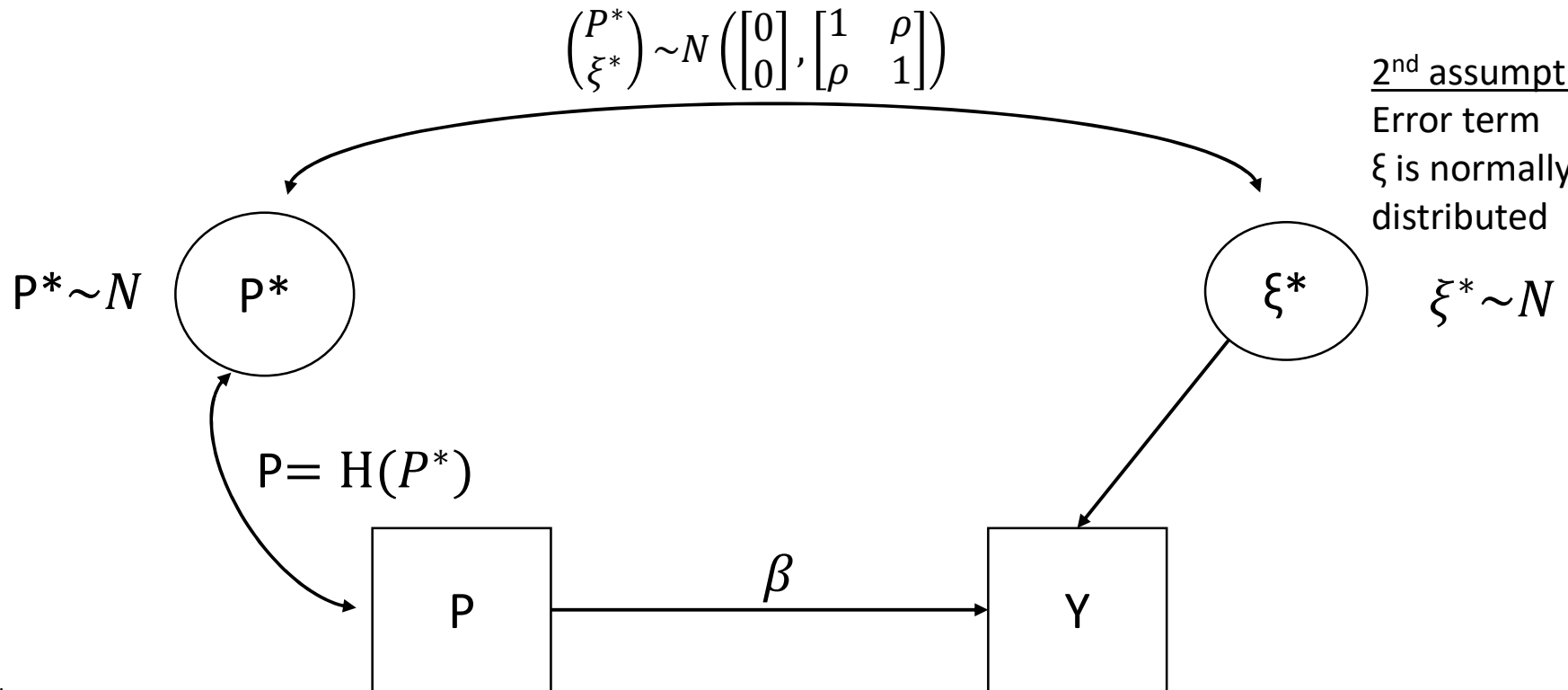
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1st assumption: Joint dependence is bivariate-normal, with correlation ρ , such that a Gaussian copula captures the dependence

$$\begin{pmatrix} P^* \\ \xi^* \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

2nd assumption:
Error term
 ξ is normally
distributed

$$\xi^* \sim N$$



3rd assumption:

Regressor P is sufficiently
non-normally distributed $P \not\sim N$

$$Y = \alpha + \beta P + \xi$$

$$\xi = \xi^* \sigma$$

Let's do an exercise before we continue...



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EXPRESS: A Practical Guide to Endogeneity Correction Using Copulas

Yi Qian, Anthony Koschmann, and Hui Xie [View all authors and affiliations](#)

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Table 4. Assumptions Used to Derive 2sCOPE/2sCOPE-np.

Assumptions

- Assumption 1. Either the error E_i or its endogenous part U_i is normally distributed (Double Robustness 1).
Assumption 2. Either $(P, E) | W$ or $(P, U) | W$ follows a Gaussian copula (Double Robustness 2).^a
Assumption 3. Full rank of all regressors and $\text{cov}(W, E) = 0$.^b
Assumption 4. P is continuous and the copula term $C_{p|w}$ is linearly independent of (I, P, W) .

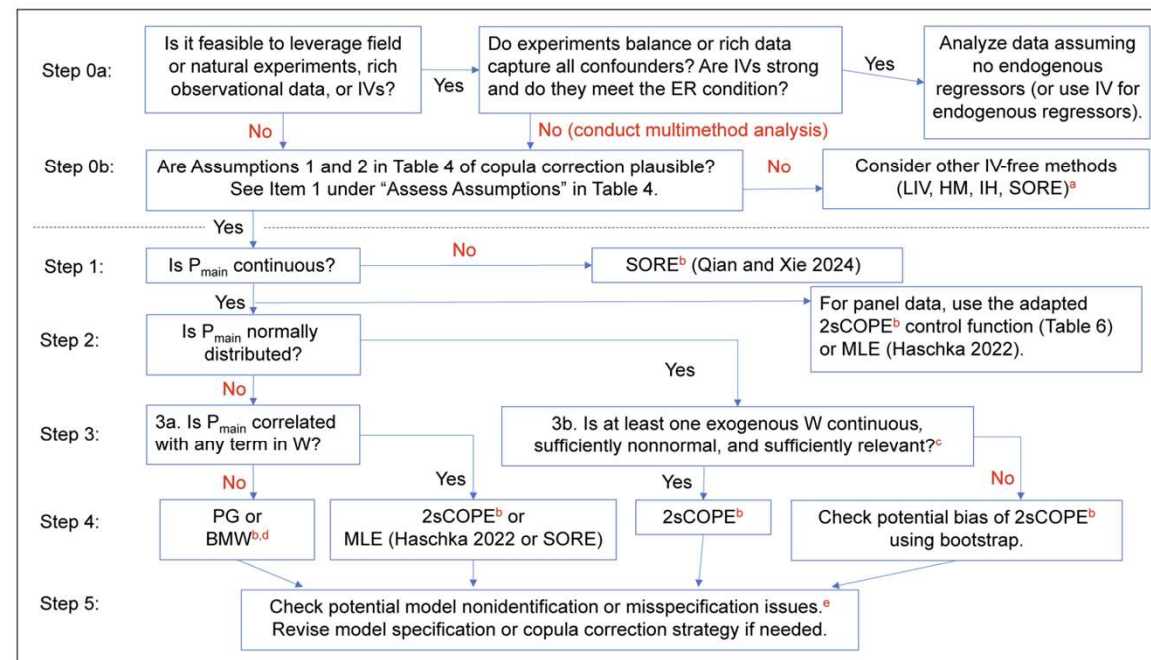


Figure 5. Flowchart for Copula Procedure.

Explicit *and implicit* assumptions for Gaussian Copula Control Function (GCCF) methods

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1st assumption: Joint dependence is bivariate-normal, with correlation ρ , such that a Gaussian copula captures the dependence

$$\begin{pmatrix} P^* \\ \xi^* \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

2nd assumption:
Error term
 ξ is normally
distributed

$$\xi^* \sim N$$

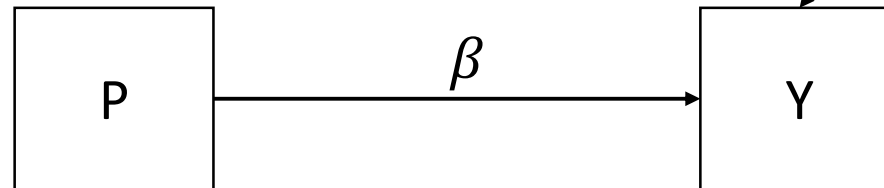
$$P^* \sim N$$

4th, an implicit assumption:
 $H()$ is a bijective and non-linear one-to-one mapping between normally distributed P^* and non-normally distributed P

$$P = H(P^*)$$

3rd assumption:

Regressor P is sufficiently non-normally distributed $P \not\sim N$



$$Y = \alpha + \beta P + \xi$$

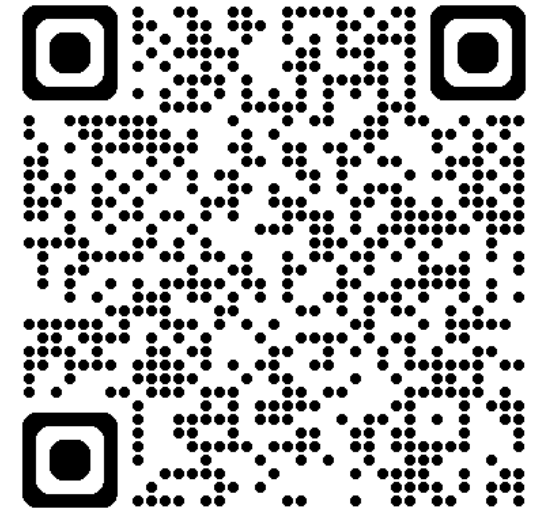
$$\xi = \xi^* \sigma$$

Notes/References: Park & Gupta (2012)

Dost & Haschka (2025) The Gaussian Copula Control Function Method does not help against Traditional Omitted Variable Bias. <https://ssrn.com/abstract=5285127>

Let's see it in action, let's power up R...

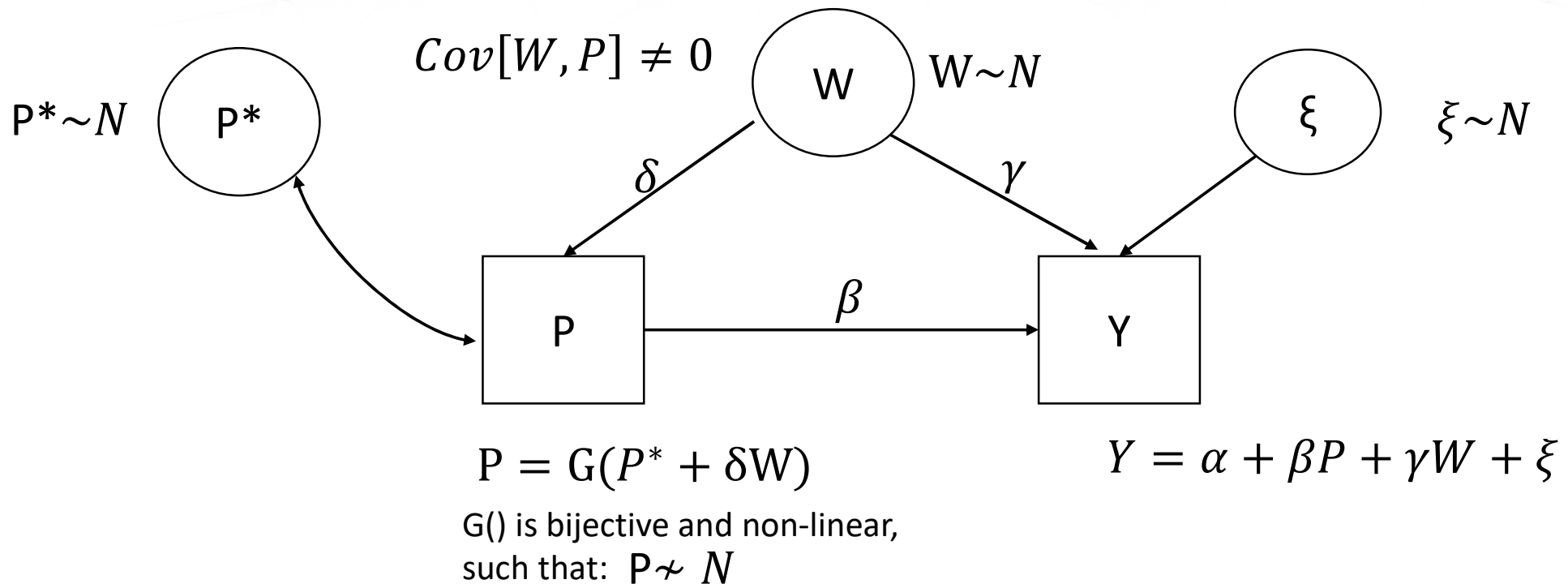
- **Materials:**
<https://github.com/HashtagHaschka/Workshop>
- **R-File:**
part_I_code.R



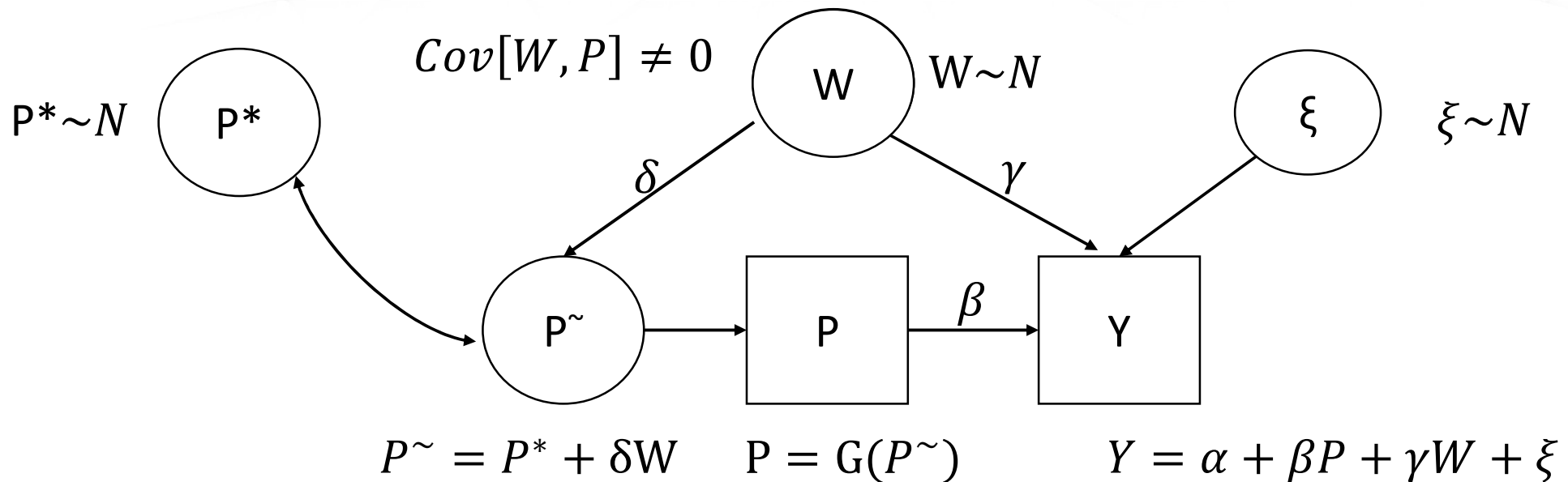
Omitted variable DGP with additive process and nonlinear transform

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Equivalent notation with intermediate step:



Necessary!

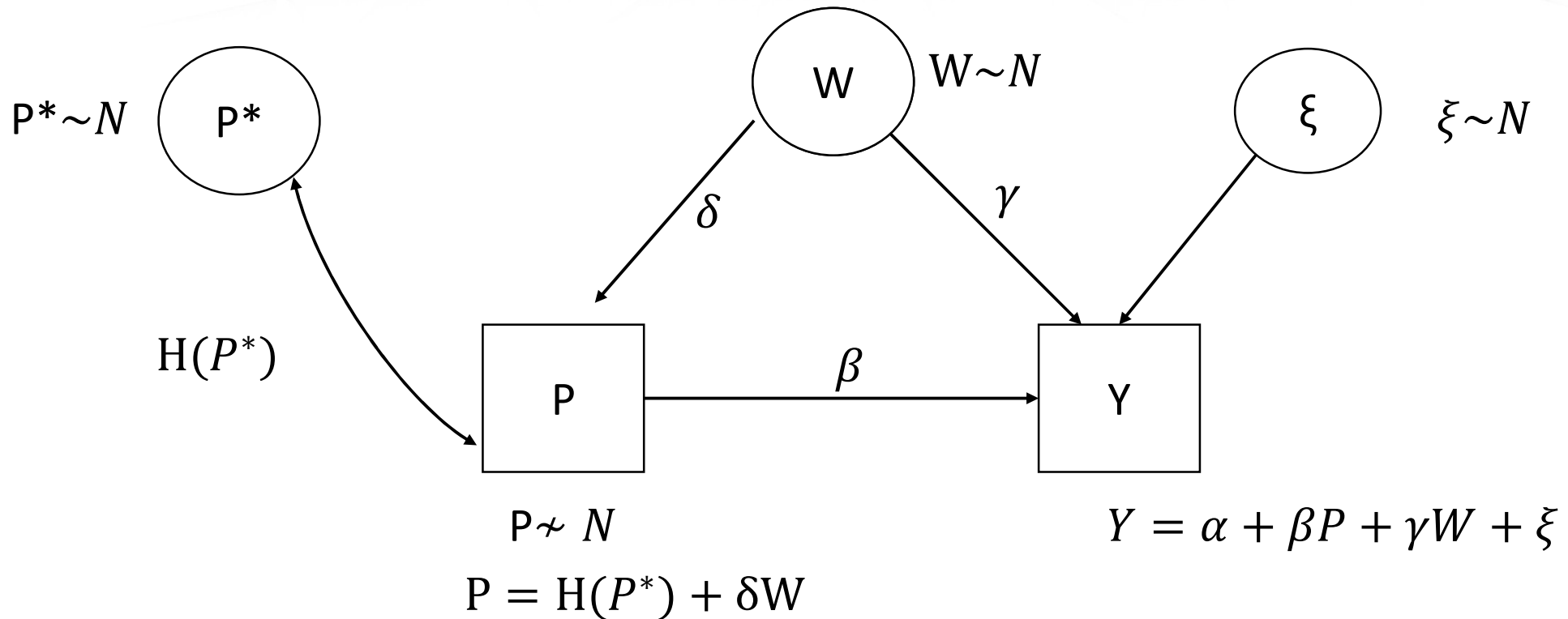
$G()$ is bijective and nonlinear

How realistic is such a DGP?

Traditional omitted variable DGP with non-Gaussian exogenous component

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Prove that GCCF estimation
creates a bias:

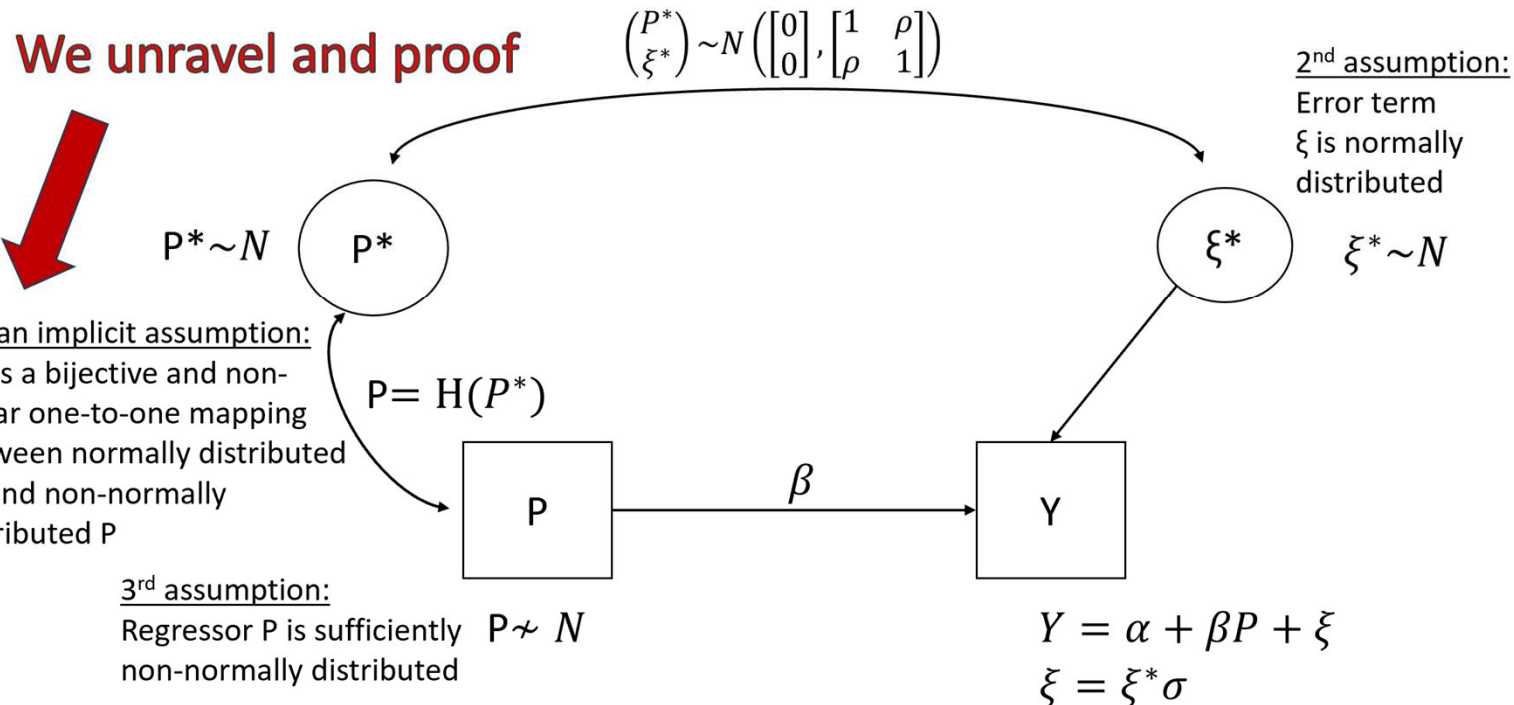
$$\text{Bias} = \gamma \cdot \frac{\delta \sigma_W^2 - \text{Cov} \left(\mathbb{E}[P_t | \tilde{P}^*], \mathbb{E}[W_t | \tilde{P}^*] \right)}{\mathbb{E}[\text{Var}(P_t | \tilde{P}^*)]}$$

The problem: Bijectivity of H rules out additive decomposability!

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1st assumption: Joint dependence is bivariate-normal, with correlation ρ , such that a Gaussian copula captures the dependence

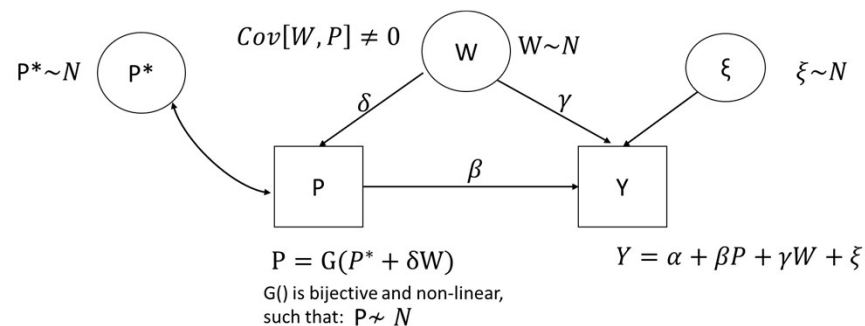


- Since H is bijective, $P^* = H^{-1}(P)$, P and P^* contain exactly the same information
- The model has only one latent dimension
- A bijective transformation of a scalar latent variable **cannot be uniquely decomposed into separate additive components**

Two simulation studies for illustration

Scenario A:

Omitted variable with bijective transform

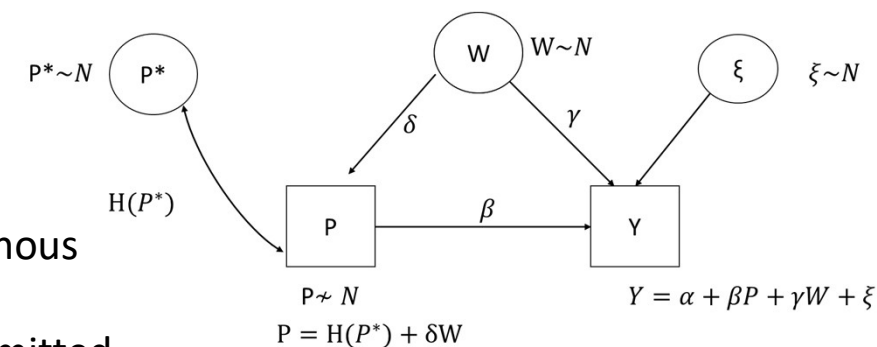


$P_t = G(P_t^* + \delta W_t)$ is the observed variable

- $Y_t = \alpha + \beta P_t + \gamma W_t + \xi_t$,
- ξ_t is structural error with $E[\xi_t | P_t, W_t] = 0$, and $\xi_t \sim N(0, 1)$,
- $\alpha = 0, \beta = 1, \gamma = 1$,
- $n = 1000$ sample size, $r = 500$ repetitions
- $P_t^* \sim N(0, 1)$ is exogenous variance,
- $W_t \sim N(0, 1)$ is the omitted variable,
- δ is randomly drawn from $[0, 0.4]$ to induce endogeneity.
- $G() = H() = \Phi^{-1}()$ the canonical nonlinear transform to uniform.
- **Bias** = $\beta_{\text{true}} - \beta_{\text{OLS/GCCF}}$

Scenario B:

Traditional omitted variable



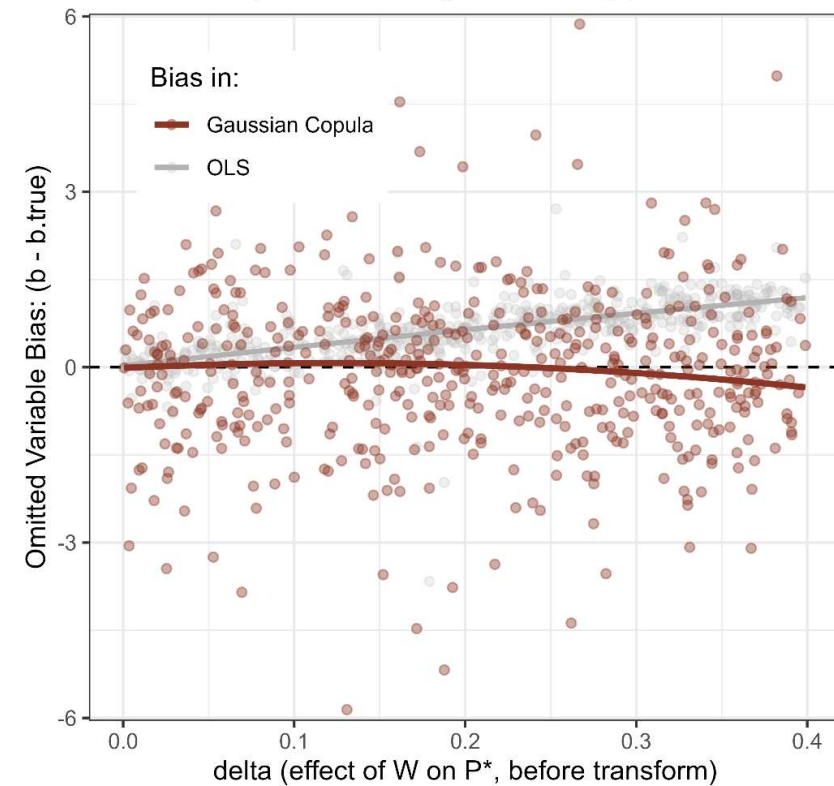
$P_t = H(P_t^*) + \delta W_t$ is the observed variable

Simulation studies results:

Scenario A:

Omitted variable with bijective transform

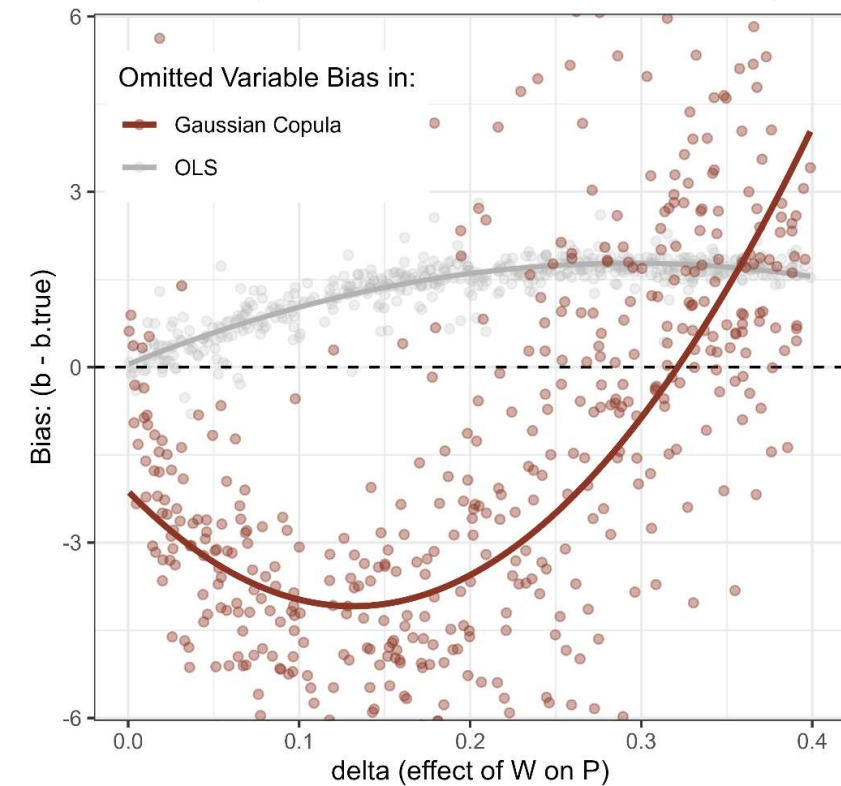
ScenarioA (transform: $G(P^* + \delta W)$)



Scenario B:

Traditional omitted variable

ScenarioB (traditional additive omitted variable)



A quick summary

The common definition of omitted variables settings:

$$Y = \alpha + \beta P + \gamma W + \xi$$
$$P = \eta + \delta W + v$$

- Additive and linear relationship between P and W
- But the Gaussian copula endogeneity correction is extra *biased* in this scenario...

The Gaussian copula endogeneity correction requires the DGP to be:

$$P = G(P^* + W), P^* \sim N, G \text{ is bijective}$$

Recommendations for GCCF use:

TLDR: Explicitly answer: Why is P **endogenous**? **AND:** Why is P **non-normal**?

- **1. Diagnose the DGP first, estimate second.** Begin every empirical project by writing down a verbal or formal data-generating process. If the story is “unobserved trait W adds to both P and the error,” copula correction is contraindicated.
- **2. Treat marginal non-normality as necessary, never sufficient.** Confirming that P is non-normal merely clears only one of several hurdles; it does not establish that a monotone transform of the endogenous component drives P .
- **3. Document the monotone-transform rationale.** When using any copula method, spell out why the underlying mechanism should be nonlinear and strictly monotone (e.g., saturation, diminishing returns, ranking processes).
- **4. Document the order of additions and transforms in the DGP.** Importantly, the transform needs to happen *after* the omitted variable additively affected the exogenous variance in P .
- **5. Triangulate with IVs when possible.** If a credible instrument is available, compare IV and GCCF estimates. Divergence is an immediate red flag that the monotone-transform assumption may be violated.

Are these correct? (Qian, Koschmann, Xie, 2026, JM)

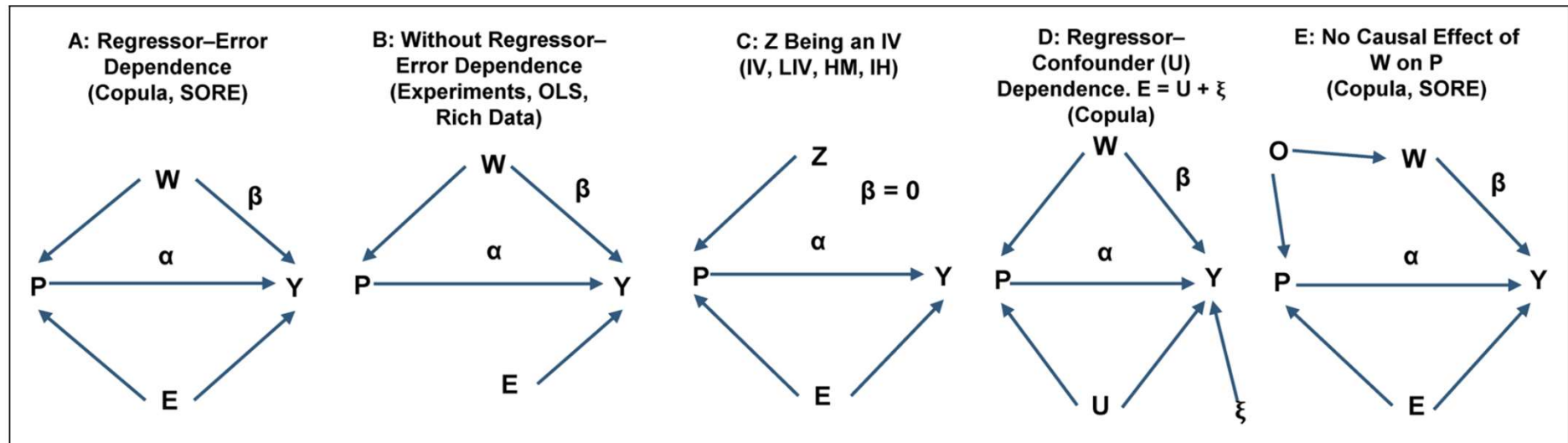


Figure 2. Directed Acyclic Graphs for Endogeneity.

Notes: OLS = ordinary least squares.

Strictly speaking, yes, BUT: here P has NO other exogenous variance; it is fully determined by W, U.
(see case C: here Copula is not recommended $\rightarrow$ this is the traditional, textbook linear model case)

How to spot „additive DGP language“

Synthetic Formulation of Rationales	The Structural Mismatch with GCCF	Relative Frequency
Extensive Covariates <i>“To minimize omitted variable bias, we include an extensive set of control variables.”</i>	Additive Controls vs. Non-linear Residuals: Modeling confounders as additive factors while assuming the residual error requires a non-linear correction creates a bifurcated, inconsistent DGP.	Very High
Absorbing Heterogeneity <i>“To control for unobserved heterogeneity, we include a set of firm, industry, and year fixed effects.”</i>	Additive Shifts vs. Non-linear Copulas: Fixed effects treat unobservables as additive linear intercept shifts. Using GCCF for the remaining error assumes a conflicting non-linear structure for the same DGP.	Very High
Mitigating Endogeneity <i>“The inclusion of fixed effects resolves potential endogeneity issues arising from time-invariant omitted variables.”</i>	Separability vs. Joint Dependence: If endogeneity is “resolved” by additive subtraction, the non-linear dependence required for GCCF identification is theoretically absent.	High
Dimensionality Reduction <i>“We apply PCA to our set of controls to reduce dimensionality.”</i>	Linear Redundancy vs. Non-linear Dependence: PCA assumes that entering variables are linearly related. This contradicts the GCCF requirement that unobservables must have a non-linear dependence with observables.	Moderate
Transforming Data <i>“We employ first-differencing to eliminate persistent unobserved factors from the model residual.”</i>	Linear Cancellation vs. Non-linear Persistence: Linear transformations to cancel unobservables assume a purely additive structure, whereas GCCF requires a non-linear relationship.	Low

My name is Florian, I have an omitted variables problem...

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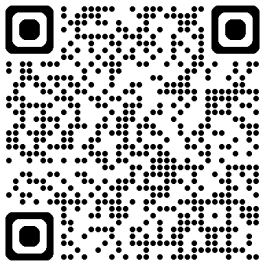
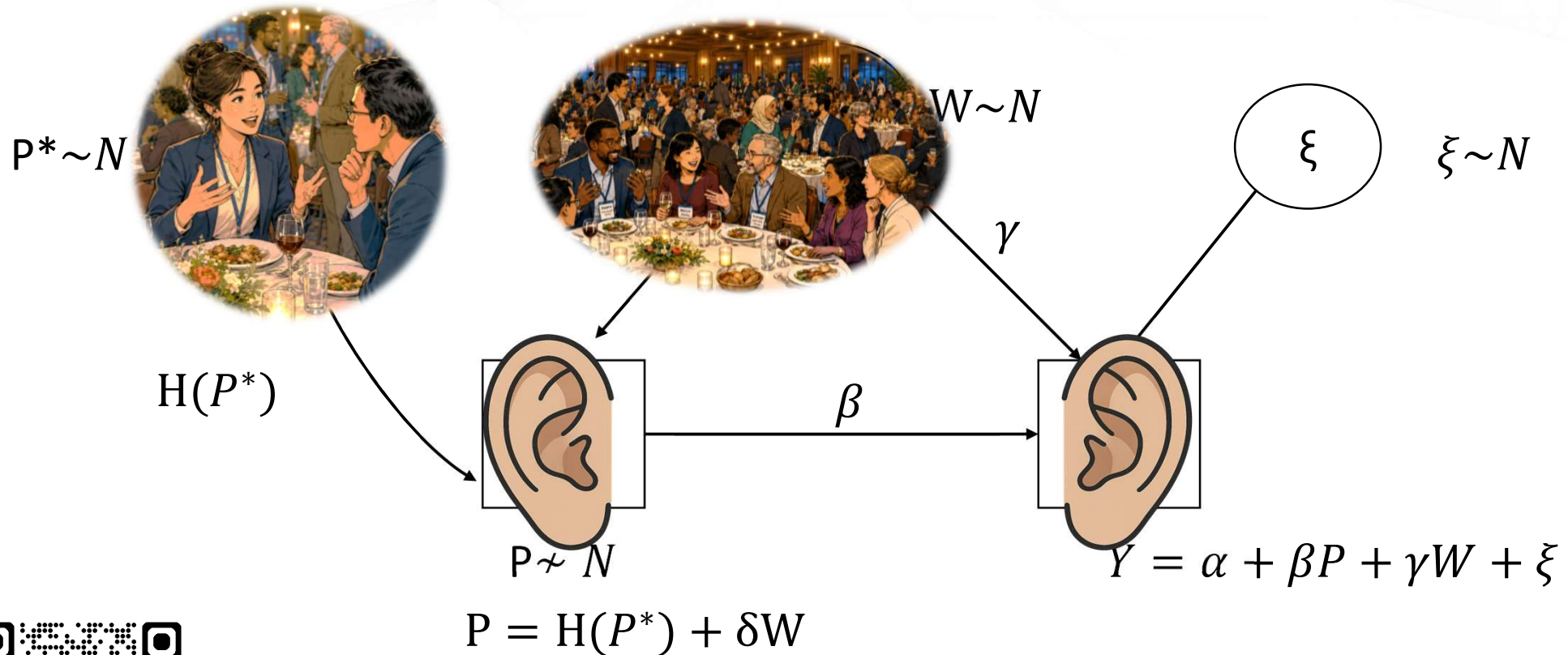
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- Criticism without solution is unfair...
- Now, what to do? We need a new method idea...

Reframe the omitted variable as “the cocktail party problem” from signal processing

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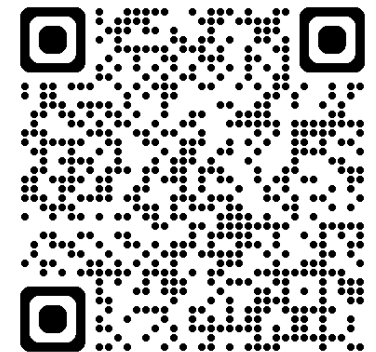
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- Implementation of an estimate for W as control function (R code)
- Extensive simulation evidence for robustness in wide array of DGPs
- canonical empirical cases

Code and paper (with identification proof) b-tu

```
15 # Step 3: Apply Independent Component Analysis (ICA)
16 data_matrix <- cbind(Y, P)
17 ica_result <- ica(X = data_matrix, nc = 2, method = "jade")
18 # Step 4: Find the ICA component that looks most normally
    distributed
19 hist(ica_result$S[, 1])
20 hist(ica_result$S[, 2])
21 # Step 5: Estimate the regression model, controlling for the normal
    component
22 model <- lm(Y ~ P + ica_result$S[, 1])
23 # View the results
24 summary(model)
```

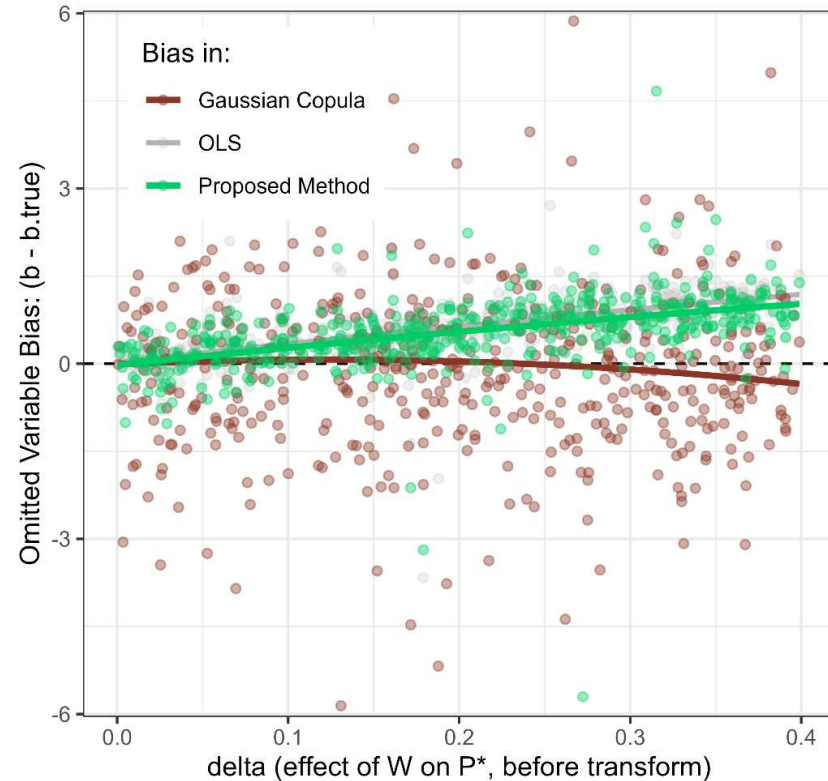


Revisit simulation studies:

Scenario A:

Omitted variable with bijective transform

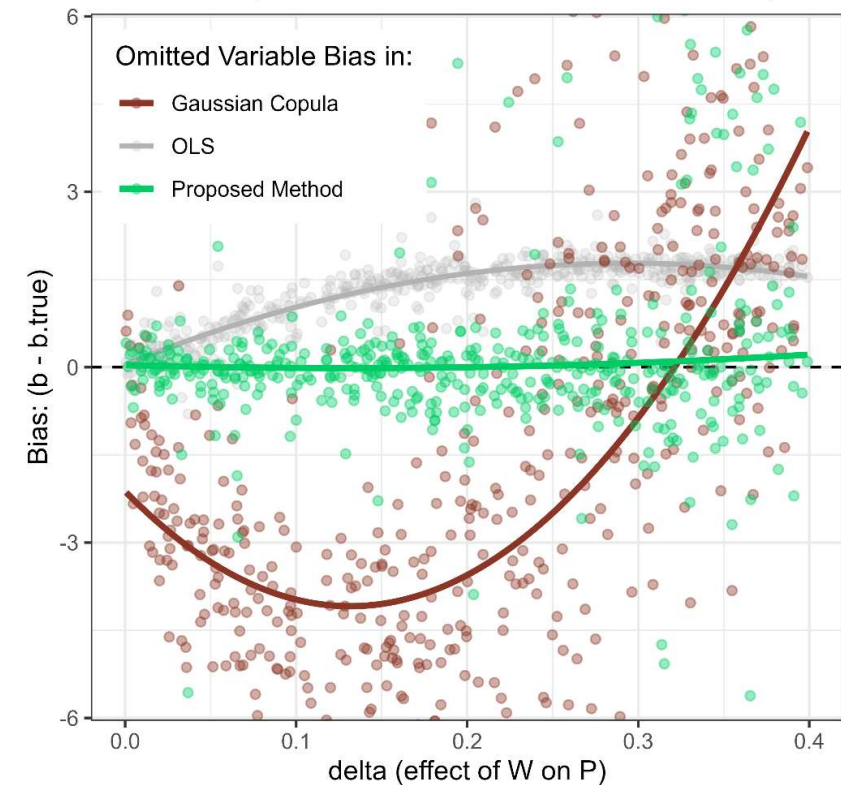
ScenarioA (transform: $G(P^* + \delta W)$)



Scenario B:

Traditional omitted variable

ScenarioB (traditional additive omitted variable)



Using Independent Component Analysis (ICA) to disentangle exogeneous from omitted signal

- ICA:
 - Blind source separation technique
 - Based on whitening and rotation
 - Several variants (e.g., JADE)
- ICA assumptions:
 - As many (or more) measured signals as independent latent sources
 - Each signal is a linear combination of sources
 - Ideally, all sources are uncorrelated and non-normal
 - At most one source can be normal
- Additional assumptions for our use case:
 - The exogeneous signal is non normal; the omitted signal (source of endogeneity) is normal
→ ICA is blind, we need the assumption to decide which component is exogenous

Using Independent Component Analysis (ICA) to disentangle exogeneous from omitted signal

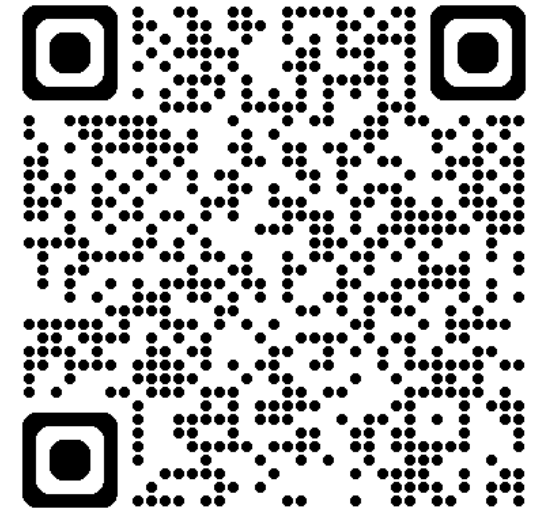
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- Additional assumptions for our use case:
 - The exogeneous signal is non normal; the omitted signal (source of endogeneity) is normal
→ ICA is blind, we need the assumption to decide which component is exogenous

Identification requires (4 assumptions):

1. Endogenous regressor can be additively decomposed into
 - Exogenous non-normal part, and
 - Endogenous normal part
2. Structural error is normal (or decomposable and endogenous part normal)
3. Endogenous normal part enters regressor and error linearly
4. Full rank of regressors and conditional exogeneity of exogenous regressors

Let's see it in action, let's power up R...

- **Materials:**
<https://github.com/HashtagHaschka/Workshop>
- **R-File:**
part_II_code.R



A true omitted variables scenario

DGP:

- $Y = \alpha + \beta P + \gamma \sum_{j=1}^J W_j + \xi, \quad \xi \sim N(0,1)$
- $P = \eta + \delta \sum_{j=1}^J W_j, \quad \eta \sim Ga(1,1)$
- We do not observe W and regress Y on P

Omitted variables:

- $J \in \{30, 25, \dots, 1\}$ latent variables W_j
- Drawn from various distributions: Uniform, Exponential, Binomial, Poisson, Gamma, Lognormal, Weibull, Beta, etc.
- Variables are summed and scaled by $1/\sqrt{J}$ to ensure unit variance (for comparability)
- None of the individual omitted variables is normally distributed
- According to CTL: their sum should be normal. This is what we exploit!

Two scenarios:

- Independent omitted variables
- Correlated omitted variables (pairwise correlations of 0.5).

A true omitted variables scenario, results

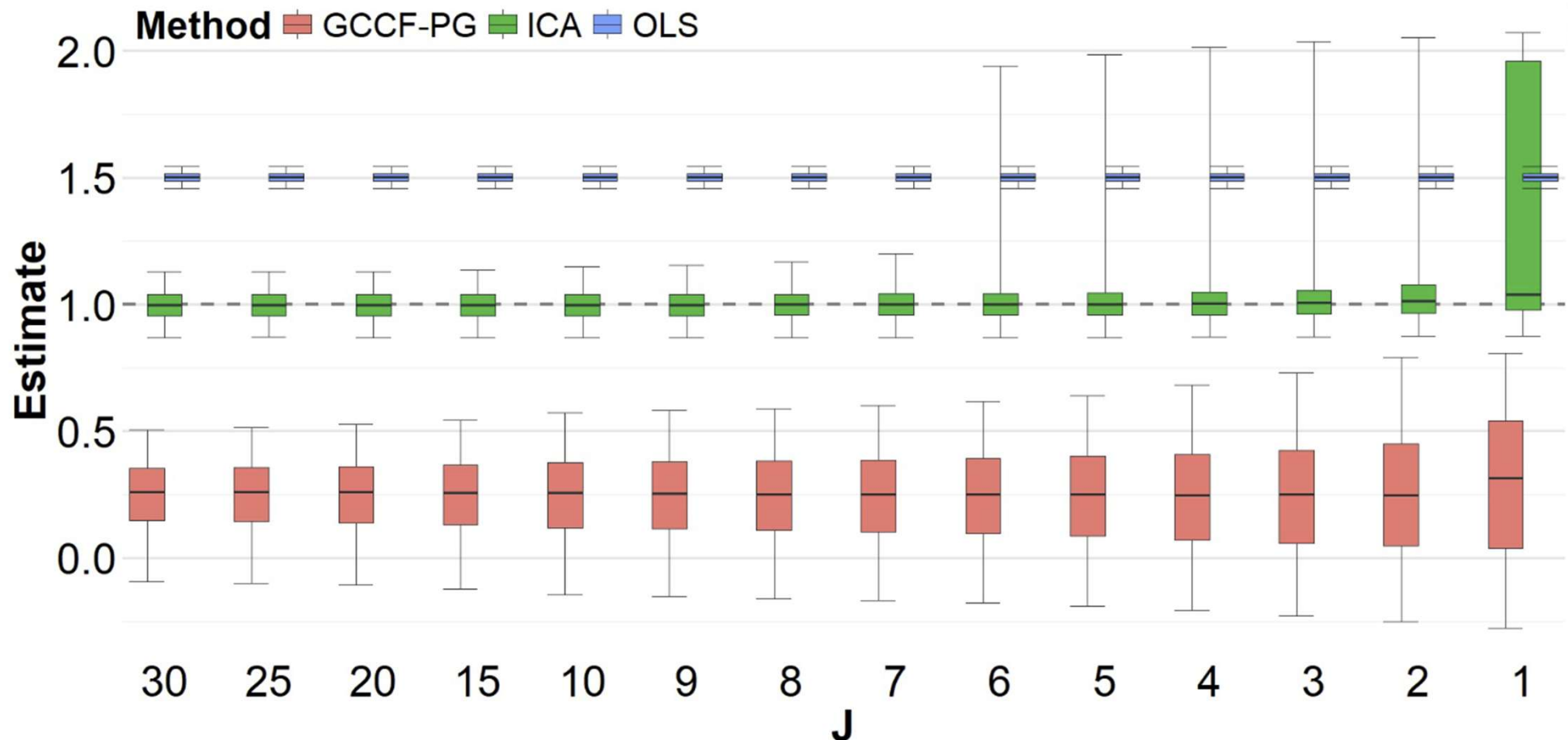


Figure 8: The boxplots show the distribution of $\hat{\beta}$ against the number of omitted variables (J). The true coefficient is $\beta = 1$ (dashed grey line). The sample size is $N = 1,000$ and the number of replications is $R = 100,000$.

Empirical study: Minute Maid orange juice price elasticity (Dominicks data)

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- Unbalanced panel (N = 9,649)
 - 8 brands
 - 83 stores
 - Week 24 1990 to week 40 1992
- Controls:
 - Quarterly dummies
 - Store Fes
 - Display dummy
 - Premium brands price
 - National brands price
 - Store brand price
- IV:
 - Frozen orange juice futures (CE FCOJ-A futures)

$$\log(sales_{it}) = \alpha + \beta_p \log(price_{it}) + X'_{it}\beta_x + \xi_{it},$$

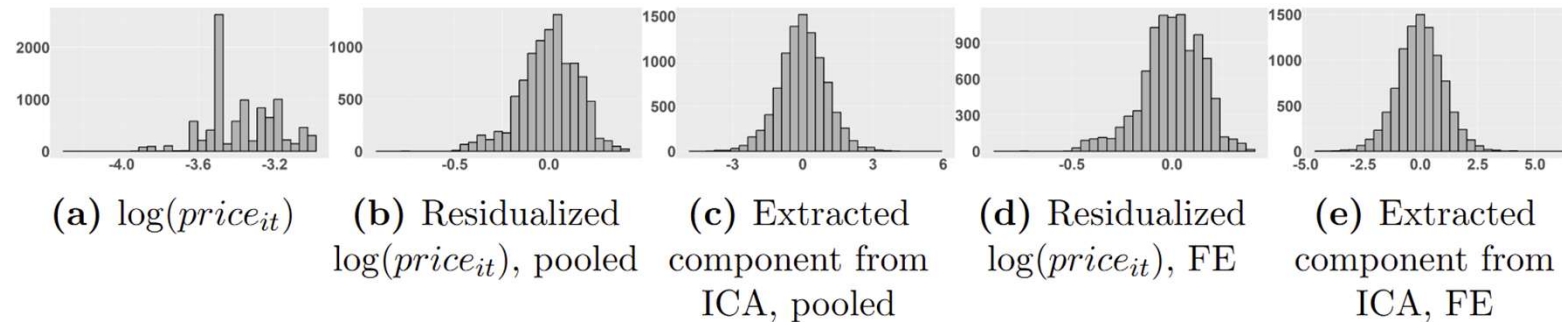
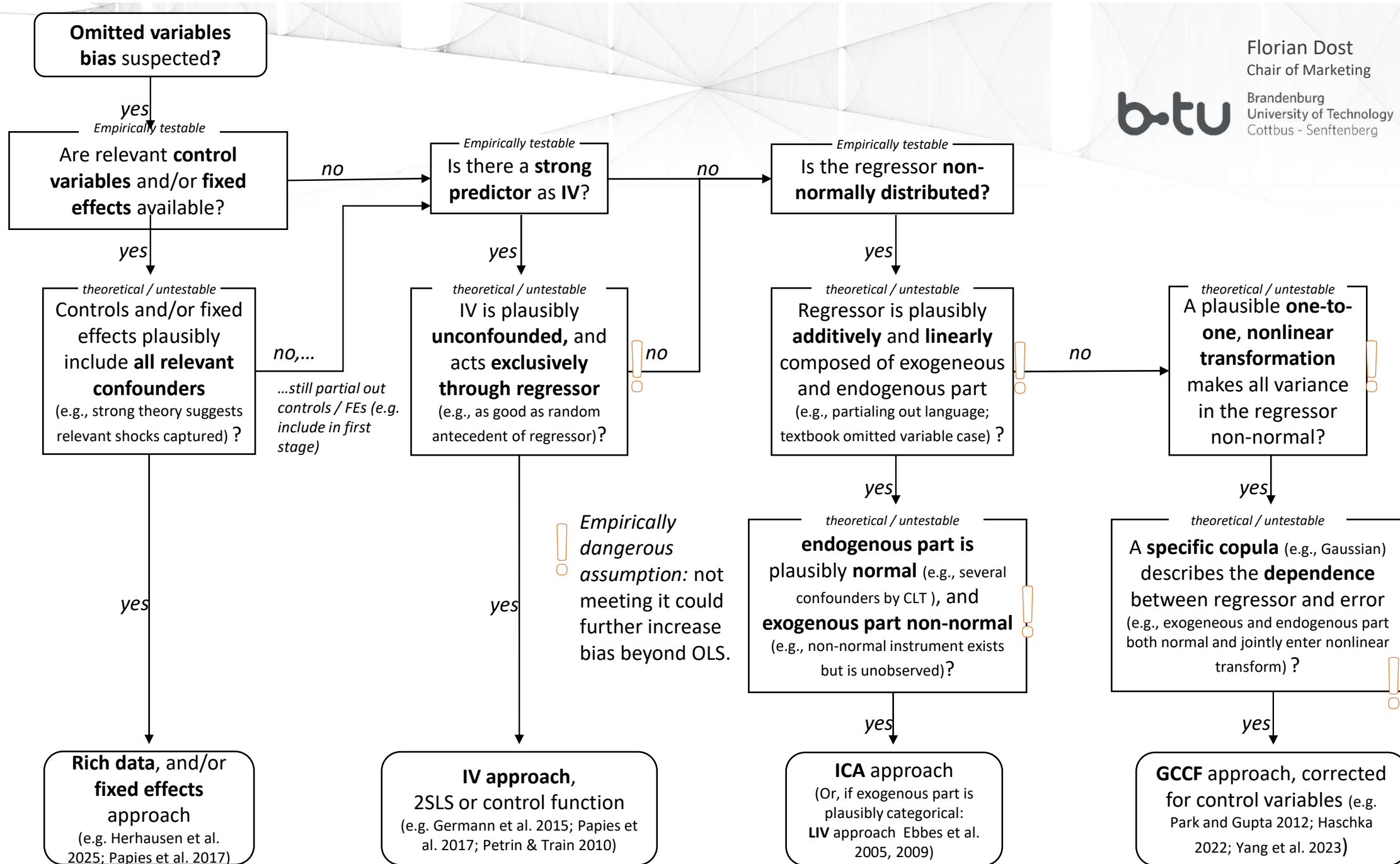


Figure 10: Histograms for the fixed-effects model (FE) and the pooled model (pooled).

Empirical results

	OLS		IV		2sCOPE-GC		Panel-GC	Proposed ICA	
Store fixed effects	No	Yes	No	Yes	No	Yes	Yes	No	Yes
Intercept	7.083 (.1885)	6.614 (.1609)	6.423 (.2278)	5.813 (.2043)	7.215 (.2148)	6.871 (.1977)	-.0067 (.0055)	6.499 (.2370)	5.909 (.2141)
<i>log price</i>	-2.126 (.0391)	-2.135 (.0319)	-2.778 (.0973)	-2.776 (.0765)	-2.005 (.1078)	-1.953 (.0920)	-3.433 (.1334)	-2.703 (.1435)	-2.700 (.1130)
<i>log price_premium</i>	.5183 (.0396)	.5269 (.0316)	.7802 (.0563)	.7676 (.0444)	.4721 (.0570)	.4638 (.0445)	.9877 (.0551)	.7500 (.0686)	.7390 (.0507)
<i>log price_national</i>	.3827 (.0356)	.2967 (.0286)	.4934 (.0424)	.3868 (.0345)	.3611 (.0391)	.2709 (.0299)	.5110 (.0360)	.4806 (.0412)	.3761 (.0303)
<i>log price_Dominicks</i>	.5951 (.0259)	.5730 (.0207)	.6733 (.0307)	.6376 (.0249)	.5823 (.0282)	.5576 (.0222)	.6889 (.0257)	.6643 (.0318)	.6299 (.0249)
<i>display</i>	1.259 (.0159)	1.253 (.0128)	1.129 (.0248)	1.125 (.0200)	1.283 (.0262)	1.289 (.0221)	1.068 (.0296)	1.144 (.0329)	1.140 (.0262)
<i>display_premium</i>	.0843 (.0135)	.0797 (.0108)	.0975 (.0151)	.0888 (.0125)	.0823 (.0134)	.0781 (.0106)	.0930 (.0083)	.0959 (.0140)	.0877 (.0113)
<i>display_national</i>	-.1894 (.0143)	-.1836 (.0114)	-.1661 (.0162)	-.1582 (.0135)	-.1938 (.0150)	-.1912 (.0118)	-.2089 (.0104)	-.1688 (.0157)	-.1612 (.0126)
<i>display_Dominicks</i>	-.1291 (.0144)	-.1251 (.0114)	-.1235 (.0160)	-.1198 (.0133)	-.1299 (.0146)	-.1262 (.0117)	-.2019 (.0114)	-.1241 (.0156)	-.1204 (.0127)
Quarterly dummies	Yes		Yes		Yes		Yes	Yes	

Notes/References:



Reviewer perspective: what to ask?

Reviewers

- **Are estimand and main threats well defined and plausible?**
 - Consider coherence between estimand and known antecedents and consequences
 - Suggest obvious and plausible unmentioned confounders
- **Are available controls used, and used controls relevant?**
- **After prevention, does an omitted-variables threat plausibly remain?**
 - When convinced of unbiased estimation, suggest sensitivity or threshold test, without extra correction
- **Is the regressor plausibly and demonstratively non-normal?**
 - When in doubt, ask for multiple common normality test statistics and a histogram
 - Check if non-normality persists *after* partialling out relevant controls (they could correlate with regressor)
- **Is the DGP plausibly additive?**
 - This is the textbook default
 - Look for tell-tale signs such as use of controls, fixed effects, or PCA
- **Is the confounder plausibly normal?**
 - Consider known distributions in case of well-established confounders in extant studies
 - Several confounders in an additive DGP will always converge on a normal distribution due to the CLT
- **How is (normal) endogeneity introduced while the regressor became non-normal?**
 - Consider plausible reasons why the regressor exhibits skew or thin or fat tails from an exogenous antecedent
 - In case authors suggest use of GCCF, use extra scrutiny: GCCF require a DGP with a one-to-one, nonlinear transform; even with a stated transform, with any factor added after, or on top of the transform GCCF methods will not only fail to reduce bias, but likely lead to *worse* bias than OLS. Ask: is such a transform plausible and economically motivated?
- **Is the method well implemented with replicable results?**
 - ICA-based methods require robust standard errors from a bootstrap procedure
 - Check replication package and code
- **Was it worth the extra effort?**
 - Triangulate with other results (e.g., OLS, IV-based)
 - Every correction method is less efficient than OLS if not needed and should not be used "pro forma"

The author's guide to do ICA

Steps:

1. Define estimand, main threat, and confounders
2. Ensure continuous endogenous regressor (ER) and dependent variable (DV), and sufficient (>250) obs
3. Prevent/reduce bias before cure, then regress ER on fixed effects and exogenous covariates, save residuals
4. Assess non-normality of (residual) ER: plot histogram, skewness, kurtosis; test against normal distribution (e.g. Kolgomorov-Smirnov and Jarque-Bera tests)
5. Argue for additive DGP (which is the traditional textbook case) with normal confounder (e.g., using CLT), and non-normal exogenous variance (e.g., given if a non-normal IV or antecedent exists)
6. Run two-components ICA on ER and DV using JADE, compute the excess kurtosis of both components, then save the more normal-looking one (smaller absolute excess kurtosis) as control function term
7. Run target model, including prevention (covariates, fixed effects) and control function term, Jackknife standard errors, report results and diagnostics

Hypothetical example as text template:

“We study the effect of review ratings (a continuous variable), as proxy for product quality [P], on sales [Y]. A threat is omitted-variable bias arising from unobserved strategic behaviors such as managers pushing buyers for better ratings, and fake reviews (both harmful and beneficial), from manufacturer, seller, and possibly competitors [*confounders*]. Even with extensive controls and weekly fixed effects [*prevention*], residual confounding may remain. Unfortunately, we have no strong instrument available; yet the residualized review ratings display a non-normal distribution: (*skewness* = ..., *kurtosis* = ..., Jarque-Bera p = ..., Kolmogorov-Smirnov p = ...) [*non – normality*], allowing us to consider an instrument-free ICA method. Identification requires that the confounder(s) be approximately normal and, together with non-normal exogenous variation, additively determine the regressor. We assume reviewers’ baseline product-quality assessments are independent, with managerial pressures entering additively. Both managerially affected and unaffected reviews are present. We also expect that our controls ‘partial out’ unobserved heterogeneity. All arguments motivate an additive DGP. Further assuming several confounding influences supports the idea of an overall approximately normal endogenous variance in the regressor. It is plausible, that the exogenous part reflecting perceived product quality is not normal for three reasons: effective regulation will affect the lower end, nonlinearly increasing cost the higher end of the quality distribution, leading to skew; reviewers perceptions and consequent ratings, may emphasize particularly negative or positive judgments, leading to excess kurtosis. We recover the normal component via JADE, then include the more normal component in the outcome equation as control function. Inference uses Jackknife standard errors. Results are as follows...”

BEWARE:

Already known issues with the estimator (1/2)

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Identification requires (4 assumptions):

1. Endogenous regressor can be additively decomposed into
 - Exogenous non-normal part, and
 - Endogenous normal part
2. Structural error is normal (or decomposable and endogenous part is normal)
3. Endogenous normal part enters regressor and error linearly
4. Full rank of regressors and conditional exogeneity of exogenous regressors

Potential issues in empirical cases	Method may break down, because...	Indications
<i>Related to assumption 1 (exogenous non-normal and endogenous normal part):</i>		
A single, or dominant non-normal OV	Researcher may choose the wrong ICA component as control term	ICA outputs two non-normal components
Normal (residual) endogenous variable	ICA cannot separate several normal components	Endogenous variable looks normal (e.g., insignificant KS test)
Normal (measurement) noise in endogenous variable	May inflate standard errors	Abnormally large standard errors
Confounder enters the regressor non-linearly	Endogenous component becomes non-normal, so ICA may select a distorted component	ICA outputs two non-normal components (largely robust unless distortion is strong)
<i>Related to assumption 2 (Error is normal in endogenous part):</i>		
Skewed or non-normal error	ICA may identify distorted components	(Method appears robust to violations)
Normal noise in error/ measurement noise in DV	May slightly inflate standard errors	Abnormally large standard errors
Non-normal noise in error	Two non-normal sources compete	No ICA component has near-zero excess kurtosis

BEWARE:

Already known issues with the estimator (2/2)

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Identification requires (4 assumptions):

1. Endogenous regressor can be additively decomposed into
 - Exogenous non-normal part, and
 - Endogenous normal part
2. Structural error is normal (or decomposable and endogenous part is normal)
3. Endogenous normal part enters regressor and error linearly
4. Full rank of regressors and conditional exogeneity of exogenous regressors

Potential issues in empirical cases	Method may break down, because...	Indications
<i>Related to assumption 3 (Linear addition of OV to regressor and error):</i>		
Non-linear dependence, as in GCCF DGP	ICA may identify distorted components	(Method appears robust to violations)
Discrete endogenous or dependent variable	Link function instead of additive error in generalized models (logit regression, ordered probit, Poisson regression, etc.)	Endogenous or dependent variable discrete (note: exogenous component can be discrete, as in LIV)
A moderator shifts the regressor's slope (slope endogeneity or effect heterogeneity)	OV enters non-additively (multiplicatively), so ICA cannot recover a single slope and identification is lost	Abnormally large standard errors
<i>Related to assumption 4 (Regressors have full rank; conditional exogeneity of exogenous regressors)</i>		
Autocorrelation / heteroskedasticity in exogenous component and error	May inflate standard errors	Large standard errors; large autocorrelation in variable (e.g., PACF)
Exogenous component correlated with OV	Breaks exogeneity, leads to bias (still less than OLS)	(untestable)

Let's have a discussion!



The Gaussian copula method cannot
correct for omitted variables bias



An ICA-based alternative to correct for
omitted-variable bias without instruments

